

## Regression

### Overview

*No Class Meeting (MLK Day)*

- a **machine learning** task where we **predict** a real-valued **output** response given an **input vector**
- **linear** and **non-linear** regression **models**
- **analytic** (closed-form) **solution** to learn an **optimal model** from **training data**
- **gradient descent** and **stochastic gradient** descent algorithms
- model **complexity** and **overfitting**

### Readings

- <https://www.microsoft.com/en-us/research/wp-content/uploads/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf#page=157>

### Bishop, Pattern Recognition and Machine Learning (2006) pp. 137-152

#### Key Concepts

|                                |                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------|
| <b>Linear Model</b>            | a <b>function</b> that <b>adds up weighted inputs</b>                                      |
| <b>Loss</b>                    | a <b>measure</b> of how <b>wrong predictions</b> are                                       |
| <b>Least Squares</b>           | find <b>weights</b> that make <b>loss</b> as <b>small</b> as <b>possible</b>               |
| <b>Basis Function</b>          | transforms <b>input</b> so <b>models</b> can <b>capture</b> more <b>complex patterns</b>   |
| <b>Maximum Likelihood</b>      | statistical way to <b>find weights</b> that make <b>observed data</b> most <b>probable</b> |
| <b>Predictive Distribution</b> | <b>range</b> of possible <b>outputs</b> with <b>associated confidence</b>                  |

#### Scope of Reading

- **fit** a linear **model** to **data**
- **measure** model **efficacy**
- **interpret** the **model** statistically

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Linear Regression Model</b>                  | <ul style="list-style-type: none"> <li>the <b>linear model</b> predicts <math>y(x) = w_0 + w_1x_1 + w_2x_2 + \dots + w_Dx_D</math><br/> where: <ul style="list-style-type: none"> <li><math>x = [x_1, x_2, \dots, x_D]</math> is a <b>vector</b> of <b>inputs</b>.</li> <li><math>w = (w_0, w_1, \dots, w_D)</math> are <b>parameters (weights)</b>.</li> </ul> </li> <li><math>w_0</math> is the <b>bias</b> (or <b>intercept</b>)</li> <li>each <math>w_j</math> <b>scales</b> the <b>corresponding</b> input <b>feature</b></li> </ul>                                                                                                                                           |
| <b>Loss Function: Sum of Squared Errors</b>     | <ul style="list-style-type: none"> <li>to <b>train</b> the <b>model</b>, we choose <b>weights</b> that make <b>predictions close</b> to real <b>targets</b></li> <li><b>measure</b> each <b>data point</b> <ul style="list-style-type: none"> <li><b>error</b> = <b>predicted value</b> – <b>actual value</b></li> </ul> </li> <li>sum <b>squared errors</b> over all <b>data points</b> <ul style="list-style-type: none"> <li><math>E(w) = \sum_{n=1}^N (y(x_n) - t_n)^2</math></li> </ul> </li> <li><b>advantages</b> <ul style="list-style-type: none"> <li>all <b>errors</b> are <b>positive</b></li> <li><b>large errors</b> are <b>penalized</b> more</li> </ul> </li> </ul> |
| <b>Finding the Best Weights (Least Squares)</b> | <ul style="list-style-type: none"> <li>set the <b>derivative</b> of <b>error</b> with <b>respect</b> to <b>weights</b> to <b>zero</b> <ul style="list-style-type: none"> <li>produces a <b>formula</b> that can be <b>solved</b> in a single <b>step</b></li> </ul> </li> <li>in <b>matrix form</b> <ul style="list-style-type: none"> <li>let <math>\Phi</math> = <b>design matrix</b> (features for all <b>data points</b>)</li> <li>let <math>t</math> = <b>vector</b> of <b>target values</b></li> <li><b>optimal weights</b>: <math>w^* = (\Phi^T \Phi)^{-1} \Phi^T t</math></li> </ul> </li> </ul>                                                                            |
| <b>Geometric Interpretation</b>                 | <ul style="list-style-type: none"> <li>the <b>model</b> attempts to find the <b>closest fit</b> of a <b>hyperplane</b> to the <b>data points</b></li> <li>the <b>error vector</b> is <b>orthogonal</b> to the <b>column space</b> of the <b>design matrix</b></li> <li>we project <b>target values</b> onto the <b>space</b> spanned by <b>basis functions</b></li> </ul>                                                                                                                                                                                                                                                                                                           |

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| <b>Basis Functions</b>                | <ul style="list-style-type: none"> <li>nonlinear <b>relationships</b> can be <b>modeled linearly</b> in <b>transformed space</b> using <b>basis functions</b> <ul style="list-style-type: none"> <li><b>polynomial</b>: <math>\phi_j(x) = x^j</math></li> <li><b>Gaussian</b>: <math>\phi_j(x) = \exp\left(-\frac{(x-\mu_j)^2}{2s_j^2}\right)</math></li> </ul> </li> <li>the model becomes: <math>y(x) = \sum_j w_j \phi_j(x)</math></li> </ul>                         |
| <b>Probabilistic Interpretation</b>   | <ul style="list-style-type: none"> <li>we <b>assume</b> targets are <b>generated by model</b> plus <b>Gaussian noise</b> with <b>zero mean</b> and <b>variance <math>\sigma^2</math></b></li> <li>likely: <math>p(t_n   x_n, w, \beta) = \mathcal{N}(t_n   w^T \phi(x_n), \beta^{-1})</math></li> <li>where: <math>\beta = 1/\sigma^2</math> is the <b>noise precision</b></li> </ul> <p>(this turns <b>least squares</b> into <b>maximum likelihood</b> estimation)</p> |
| <b>Maximum Likelihood for Weights</b> | <ul style="list-style-type: none"> <li>under <b>Gaussian noise</b>, the <b>likelihood</b> of the entire <b>dataset</b> is the <b>product</b> of individual <b>Gaussian terms</b></li> <li><b>maximizing likelihood</b> is equivalent to <b>minimizing squared errors</b></li> <li><b>least-squares</b> solution = <b>maximum likelihood</b> solution</li> </ul>                                                                                                          |
| <b>Predictive Distribution</b>        | <ul style="list-style-type: none"> <li>once we <b>estimate weights</b>, predicted distribution of a new input <math>x_*</math> is <b>Gaussian</b></li> <li>mean: <math>y(x_*) = w_T \phi(x_*)</math></li> <li><b>variance</b> depends on <b>noise variance</b> and how far <math>x_*</math> is from <b>training</b> inputs</li> <li>this gives <b>prediction</b>, and <b>confidence in prediction</b></li> </ul>                                                         |
| <b>Overfitting and Regularization</b> | <ul style="list-style-type: none"> <li>if the <b>model</b> has <b>too many parameters</b> it may <b>fit noise</b> instead of the <b>true pattern</b></li> <li>to <b>reduce overfitting</b> <ul style="list-style-type: none"> <li>add a <b>penalty</b> for <b>large weights</b> (<b>regularization</b>)</li> <li>this can be <b>interpreted</b> as <b>having a prior</b> on <b>weights</b></li> </ul> </li> </ul>                                                        |

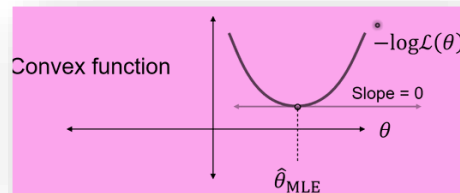
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|                          | <ul style="list-style-type: none"> <li>• <b>regularization solution</b> balances the <b>fit to data</b> and <b>simplicity</b> of <b>model</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Async</b>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Linear Regression</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Definition</b>        | <ul style="list-style-type: none"> <li>• <b>supervised learning</b> technique used to <b>train models</b> with multiple input variables mapped to a single Real value</li> <li>• we <b>map</b> the <b>vectorized inputs</b> to a corresponding <b>output</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>1-D and Multi-D</b>   | <ul style="list-style-type: none"> <li>• <b>one-dimensional</b>: models a <b>real</b> valued <b>input</b> to a <b>real</b> variable <b>response</b> so that, <math>y = mx + b</math></li> <li>• <b>multi-dimensional</b>: models <b>vectorized real</b> values to a <b>single response</b> variable, <math>\hat{y} = \beta_{i1}x_1 + \beta_{i2}x_2 + \beta_0</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>MLE Roadmap</b>       | <p><b>MLE = 'maximum likelihood estimation'</b></p> <ul style="list-style-type: none"> <li>• <i>the roadmap</i> <ol style="list-style-type: none"> <li>1. <b>establish</b> model <b>likelihood</b> for <b>data</b></li> <li>2. <b>derive</b> an analytical <b>formula</b> for optimal <b>parameters</b> on training <b>data</b></li> <li>3. <b>extend</b> to a nonlinear model</li> </ol> </li> <li>• <math>\arg \max_{\theta} \prod_{i=1}^n p\theta(y_i   x_i)</math> <b>supervised task</b>, <b>IID</b> data (<b>IID</b>: <b>Independent and Identically Distributed Data</b>; for <b>n</b> data points: <math>\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n</math>)</li> <li>• if we set up the <b>maximum likelihood estimation</b> task for linear regression will be <b>equivalent</b> for linear regression to <b>minimizing</b> the <b>sum of squared errors</b></li> <li>• <math>\arg \max_{\theta} \prod_{i=1}^n p\theta(y_i   x_i)</math> <math display="block">= \arg \max_{\theta \in \mathbb{R}^p} - \sum_{i=1}^n (y_i - \theta^T x_i)^2</math> </li> </ul> |

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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | $= \underset{\theta \in \mathbb{R}^p}{\arg \min} \sum_{i=1}^n (y_i - \theta^T x_i)^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Model and Likelihood   | <ul style="list-style-type: none"> <li>• <b>Linear Probabilistic Model:</b> <math>y = \theta^T x + \varepsilon</math>, where             <ul style="list-style-type: none"> <li>○ <math>\theta^T</math> is the <b>vector</b> of <b>parameters</b></li> <li>○ <math>x</math> is the <b>vector</b> of <b>covariances</b></li> <li>○ and <math>\varepsilon</math> is <b>real</b> value <b>noise</b> of the model: <math>\varepsilon \sim \mathcal{N}(0, \sigma^2)</math></li> </ul> </li> <li>• <b>linear</b> combination of <b>covariates</b>: <math>\sum_{i=1}^p \theta_i x_i</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Conditional Likelihood | <ul style="list-style-type: none"> <li>• <b>Conditioned on <math>x</math></b>, the response variable <math>y</math> follows a <b>Gaussian distribution</b> with mean <math>\theta^T x</math> and variance <math>\sigma^2</math></li> <li>• because we are <b>adding</b> a <b>mean zero noise</b>, and we know that <b>adding</b> a <b>constant</b> to a Gaussian is <b>equivalent</b> to <b>changing</b> its <b>mean</b>, the <b>Gaussian</b> goes from <b>zero</b> to the added <b>constant</b></li> <li>• so, <math>y</math>, given <math>x</math>, will be a <b>Gaussian centered</b> at the <b>linear prediction</b> with <b>variance</b> given by <b>sigma squared</b>, <math>Y \sim \mathcal{N}(\theta^T x, \sigma^2)</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| MLE with Dataset       | <ul style="list-style-type: none"> <li>• <b>Estimating the Model</b> <ul style="list-style-type: none"> <li>○ given data, how can we estimate <math>\theta</math>?</li> <li>○ <b>construct</b> maximum likelihood estimator (<b>MLE</b>)                     <ul style="list-style-type: none"> <li>▪ <b>derive the log-likelihood</b></li> <li>▪ <b>find <math>\theta</math> MLE that maximizes</b> likelihood                         <ul style="list-style-type: none"> <li>• <b>analytically:</b> take <b>derivative</b> and <b>set to 0</b></li> <li>• <b>iteratively:</b> (stochastic) <b>gradient descent</b></li> </ul> </li> </ul> </li> </ul> </li> <li>• <b>defining the likelihood</b> <ul style="list-style-type: none"> <li>○ <b>conditional density:</b> <math display="block">p_{\theta}(y \mid x) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right) \cdot \exp\left(-\frac{(y - \theta^T x)^2}{2\sigma^2}\right)</math> </li> <li>○ <b>likelihood function:</b> <math display="block">\mathcal{L}(\theta; \mathcal{D}) = \prod_{i=1}^n p_{\theta}(y_i \mid x_i) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\left(-\frac{(y_i - \theta^T x_i)^2}{2\sigma^2}\right) =</math> </li> </ul> </li> </ul> |

$$\frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} \exp\left(\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \theta^T x_i)^2\right)$$

### Analytical Solution

- minimizing the **likelihood** (minimizing the squared error) is a **convex function** where we **seek** the slope (**gradient**) = 0



- the **gradient** is generalizing the **derivative** for **multivariate functions**

$$-\nabla_{\theta} \log \mathcal{L}(\theta) = \nabla_{\theta} \sum_{i=1}^n (y_i - \theta^T x_i)^2$$

Chain rule →

$$= -2 \sum_{i=1}^n (y_i - \theta^T x_i) x_i$$

$$= -2 \sum_{i=1}^n y_i x_i + 2 \sum_{i=1}^n (\theta^T x_i) x_i$$

- rewriting with **matrix** notation

○ represent data  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$  as,

**covariate** (design) **matrix**:

$$\begin{matrix} n \\ \left\{ \right. \end{matrix} X = \underbrace{\begin{pmatrix} -x_1- \\ -x_2- \\ \vdots \\ -x_n- \end{pmatrix}}_p \in \mathbb{R}^{np}$$

assume  $X$  has rank  $p$  (not degenerate)

**response vector**

$$\begin{matrix} n \\ \left\{ \right. \end{matrix} Y = \underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}}_1 \in \mathbb{R}^n$$

- rewriting the **gradient** in matrix form

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|--|----------|----------|----------|--|--|----------|----------|--|--|--|----------|-------|-------|-------|-------|-------|-------|-------|-------|
|                               | $-\nabla \theta \log \mathcal{L}(\theta) = -2 \sum_{i=1}^n y_i x_i + 2 \sum_{i=1}^n (\theta^T x_i) x_i$ $= -2 X^T Y + 2 X^T X \theta$ <ul style="list-style-type: none"><li>our goal is <b>gradient = 0</b> which gives,<math display="block">X_T X \theta = X_T Y</math><math display="block">\theta = (X_T X)^{-1} (X_T Y)</math></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| Computing with MLE Parameters | <ul style="list-style-type: none"><li>computing the MLE: <math>\hat{\theta}_{MLE} = (X^T X)^{-1} X^T Y</math><ul style="list-style-type: none"><li>not typically solved by inverting <math>X_T X</math></li><li>solved using <b>direct methods</b>:<ul style="list-style-type: none"><li><b>Cholesky factorization</b><ul style="list-style-type: none"><li>up to a <b>factor of 2</b> faster</li></ul></li><li><b>QR factorization</b><ul style="list-style-type: none"><li><b>more numerically stable</b></li></ul></li></ul></li><li>solved using <b>various iterative methods</b>:<ul style="list-style-type: none"><li>(stochastic) <b>gradient descent</b></li></ul></li></ul></li></ul> <p>or use the built-in solver in your math library</p> <p>R: <code>solve(Xt %*% X, Xt %*% y)</code></p> |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| Cholesky Factorization        | <p>solve for:</p> $\hat{\theta}_{MLE} (X^T X) \hat{\theta}_{MLE} = X^T Y$ <ul style="list-style-type: none"><li>compute <b>symm. matrix</b> <math>C = X^T X</math> <math>O(np^2)</math></li><li>compute <b>vector</b> <math>d = X^T Y</math> <math>O(np)</math></li><li><b>Cholesky factorization</b><ul style="list-style-type: none"><li><b>L</b> is <b>lower triangular</b> <math>LL^T = C</math> <math>O(p^3)</math></li></ul></li><li><b>forward subs.</b> to solve: <math>Lz = d</math> <math>O(P^2)</math></li><li><b>backward subs.</b> to solve: <math>L^T \hat{\theta}_{MLE} = z</math> <math>O(P^2)</math></li></ul>                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| Solving Triangular System     | <table><tr><td><math>A_{11}</math></td><td><math>A_{12}</math></td><td><math>A_{13}</math></td><td><math>A_{14}</math></td></tr><tr><td></td><td><math>A_{22}</math></td><td><math>A_{23}</math></td><td><math>A_{24}</math></td></tr><tr><td></td><td></td><td><math>A_{33}</math></td><td><math>A_{34}</math></td></tr><tr><td></td><td></td><td></td><td><math>A_{44}</math></td></tr></table> <p>*</p> <table><tr><td><math>x_1</math></td></tr><tr><td><math>x_2</math></td></tr><tr><td><math>x_3</math></td></tr><tr><td><math>x_4</math></td></tr></table> <p>=</p> <table><tr><td><math>b_1</math></td></tr><tr><td><math>b_2</math></td></tr><tr><td><math>b_3</math></td></tr><tr><td><math>b_4</math></td></tr></table>                                                                    | $A_{11}$ | $A_{12}$ | $A_{13}$ | $A_{14}$ |  | $A_{22}$ | $A_{23}$ | $A_{24}$ |  |  | $A_{33}$ | $A_{34}$ |  |  |  | $A_{44}$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $b_1$ | $b_2$ | $b_3$ | $b_4$ |
| $A_{11}$                      | $A_{12}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $A_{13}$ | $A_{14}$ |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
|                               | $A_{22}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $A_{23}$ | $A_{24}$ |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $A_{33}$ | $A_{34}$ |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          | $A_{44}$ |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| $x_1$                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| $x_2$                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| $x_3$                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| $x_4$                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| $b_1$                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| $b_2$                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| $b_3$                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |
| $b_4$                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |          |  |          |          |          |  |  |          |          |  |  |  |          |       |       |       |       |       |       |       |       |

|             |             |             |             |
|-------------|-------------|-------------|-------------|
| $A_{11}x_1$ | $A_{12}x_2$ | $A_{13}x_3$ | $A_{14}x_4$ |
|             | $A_{22}$    | $A_{23}$    | $A_{24}$    |
|             |             | $A_{33}$    | $A_{34}$    |
|             |             |             | $A_{44}$    |

$$x_1 = \frac{b_1 - A_{12}x_2 - A_{13}x_3 - A_{14}x_4}{A_{11}}$$

$$x_2 = \frac{b_2 - A_{23}x_3 - A_{24}x_4}{A_{22}}$$

$$x_3 = \frac{b_3 - A_{34}x_4}{A_{33}}$$

$$x_4 = \frac{b_4}{A_{44}}$$

### Iterative Optimization

- **Gradient descent** is an **iterative operation** concerned with **minimizing** some **loss** with respect to certain **parameters**
- Minimize  $L(\theta)$  with respect to  $\theta$

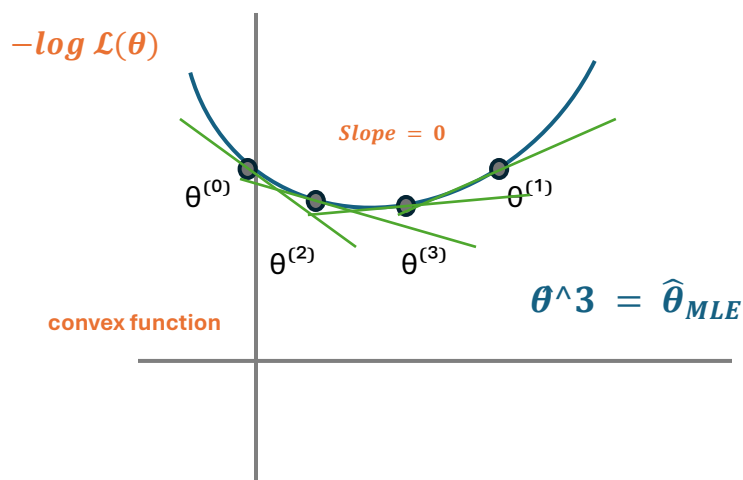
For  $\tau$  from 0 to convergence:

$$\theta^{\tau+1} = \theta^{\tau} - \underbrace{\rho(\tau)}_{\text{learning rate}} \underbrace{\nabla L(\theta^{\tau})}_{\text{direction of steepest ascent}}$$

learning rate

direction of  
steepest ascent

### Gradient Descent Illustrated





**Speeding Up  
Gradient  
Descent**

- What if  $P$  is large?

- the cost of  $O(np^2) = O(n^3)$  could be prohibitive

for  $\tau$  from 0 to convergence

$$\theta^{(t+1)} = \theta^{(t)} - \rho(\tau) \nabla_{\theta} \log \mathcal{L}(\theta^{(t)} | \mathcal{D})$$

$$\propto \theta^{(t)} - \rho(\tau) \nabla_{\theta} \sum_{i=1}^n (y_i - \theta^T x_i)^2$$

$$= \theta^{(t)} + \rho(\tau) \cdot 2 \sum_{i=1}^n (y_i - \theta^T x_i) x_i \quad O(np)$$

- can we do better?

- Solution: iterative methods

- gradient descent:

$$L(\theta) = \frac{1}{2n} \sum_{i=1}^n (y_i - \theta^T x_i)^2$$

for  $\tau$  from 0 to convergence

$$\theta^{t+1} = \theta^t - \rho(\tau) \nabla L(\theta^t)$$

$$= \theta^t + \rho(\tau) \cdot \underbrace{\frac{1}{n} \sum_{i=1}^n (y_i - \theta^T x_i) x_i}_{\text{estimate of the gradient}}$$

**Stochastic  
Gradient  
Descent**

- decomposable loss

- loss can be written as a sum of the loss on each record

$$L(\theta) = \sum_{i=1}^n L_i(\theta) = \sum_{i=1}^n L(\theta, x_i, y_i)$$

- equivalent to minimize

$$\frac{1}{n} L(\theta) = \frac{1}{n} \sum_{i=1}^n L(\theta, x_i, y_i)$$

- with gradients

$$\frac{1}{n} \nabla_{\theta} L(\theta) = \frac{1}{n} \sum_{i \in B} \nabla_{\theta} L(\theta, x_i, y_i) \approx \frac{1}{|B|} \sum_{i \in B} \nabla_{\theta} L(\theta, x_i, y_i)$$

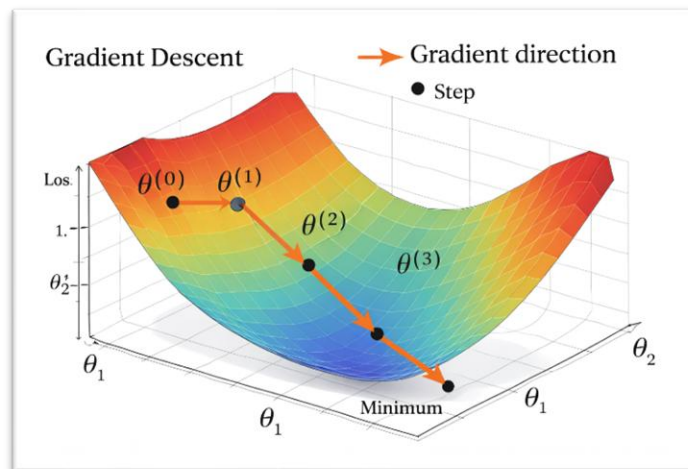
$B \sim \text{random subset of indices}$

- **decomposable loss:**

$$L(\theta) = \sum_{i=1}^n L_i(\theta) = \sum_{i=1}^n L(\theta, x_i, y_i)$$

*loss can be written as a sum of the loss on each record*

- **visualizing gradient descent**



*Image generated with ChatGPT (GPT-5.2), OpenAI. Joe Waller.*

### Stochastic Gradient Descent for Linear Regression

- for linear regression, **stochastic gradient descent** is just a matter of **randomly subsampling indices** and then **computing gradient**
- **construct a noisy estimate of the gradient:**
- for  $\tau$  from **0** to **convergence**
  1. pick a **random  $i$**
  2.  $\theta^{\tau+1} = \theta^{\tau} + \rho(\tau) (y_i - \theta^{(\tau)T} x_i) x_i$   $\mathcal{O}(np)$
- applies to **streaming data  $\mathcal{O}(p)$  storage**
- optimizing using a **single data point** at a time opens the door for **‘online’ or ‘streaming’ optimization**
- this **ability** to only consider **select data points** at a time comes in especially **useful** when **training a large model** with **limited hardware**

## Nonlinear Regression

- **linear regression** can represent nonlinear functions by **applying** nonlinear **feature transformations** and **modeling** the output as a **linear combination** of those **features**
- **representing** the design **matrix** on our chosen **feature space**,
- train:

$$\Phi(X): n \times k \quad \Phi(X) = \begin{bmatrix} -\varphi(x^1) - \\ -\varphi(x^2) - \\ \vdots \\ -\varphi(x_n) - \end{bmatrix}$$

$$\hat{\theta} \leftarrow \text{solve}_{\theta}(\Phi(X)^T \Phi(X) \theta = \Phi(X)^T Y)$$

- test:

$$x': p \times 1 \quad \varphi(x'): k \times 1$$

$$\circ \hat{y} \leftarrow \theta^T \varphi(x')$$

## Overfitting and Model Variance

- **overfitting** can give a **false sense** of doing “really well”
- **Underfitting** occurs when a machine learning **model** is too **simple** to capture the underlying **patterns** in **data**, resulting in **poor performance** on both **training** and **testing** datasets

